BIS 629 Lecture 5 Part 3: Design of stepped wedge design (SWD)-Marginal Models

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SWD Model

- We reparameterize the linear mixed model given in Lecture 3

$$Y_{ikj} = \mu_i + b_{kj} + e_{ikj}$$

with $Var(b_{kj}) = \sigma_B^2$ and $Var(e_{kj}) = \sigma_W^2$ to

$$Y_{ikj} = \beta_0 + \beta^* X_{ikj} + b_{kj} + e_{ikj} \quad (1)$$

- Then, from standard linear models theory

$$\hat{\boldsymbol{\beta}} = (X^T V^{-1} X)^{-1} (X^T V^{-1} Y),$$

and

$$Var(\hat{\boldsymbol{\beta}}) = [X^T V^{-1} X]^{-1}$$

with $V = Var(\boldsymbol{Y})$

SWD Model

- In a CRT, the X matrix is $\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \end{pmatrix}$ or $\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{pmatrix}$, where each of these matrices have dimension $Km \times 2$.
- Then, we showed in Lecture 3 part 1, (slides 26-27) that, in a balanced design, $Var(\hat{\beta}) = [X^T V^{-1} X]^{-1} = (2\sigma^2[1 + (m-1)\rho])/Km$, as the variance of the difference between the means of the clusters in two intervention groups.
- Here, we redefine K being the total number of clusters in SWD. That is, K equals the K as previously defined times 2.

SWD Model

- Following a similar approach, a cross-sectional SWD has K clusters, m_T participants/clusters/time interval, and T steps. Let Y_{jkt} be the response corresponding to individual j at time t from cluster k.
- Individual level responses in a SWD may be modelled as

$$Y_{jkt} = \beta_0 + \beta^* X_{kt} + \gamma_1 + \cdots + \gamma_{T-1} + b_k + e_{jkt}$$

- We can simplify the construction of the design matrix in a SWD by reducing the data to cluster-time level means
- Let $\bar{Y}_{kt}$ be the mean for cluster k at time t. Then

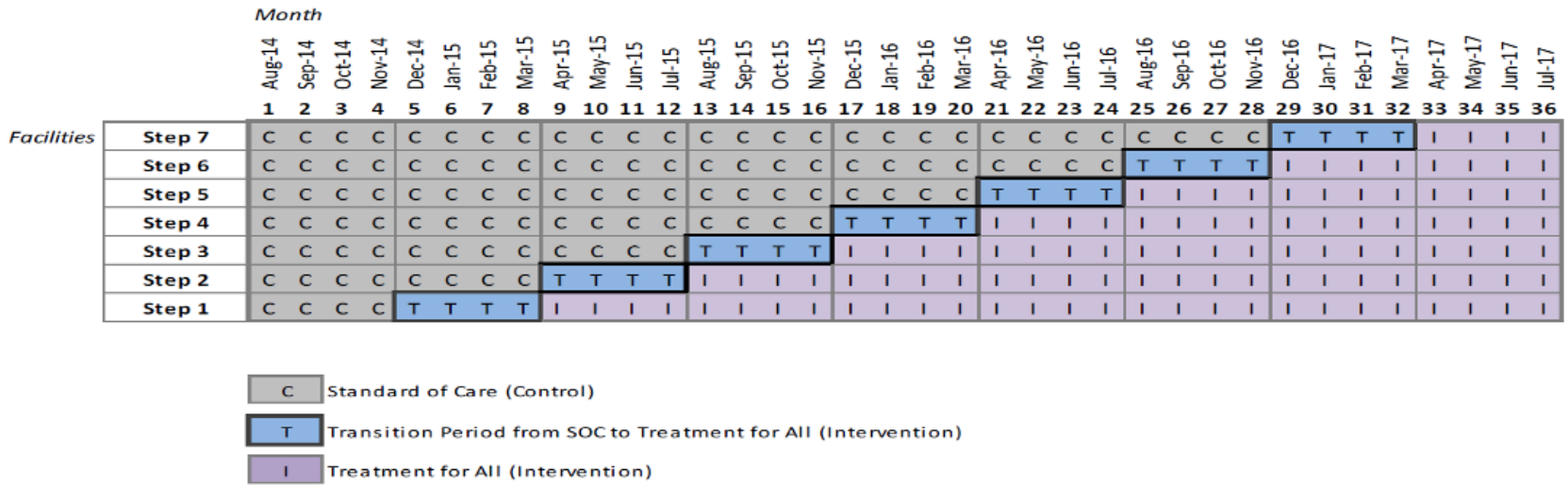
$$E_e[\bar{Y}_{kt}] = \beta_0 + \beta^* X_{kt} + \gamma_1 + \cdots + \gamma_{T-1} + b_k \tag{2}$$

where X is now a $(K \times T) \times (T + 1)$ matrix.

SWD Model

- What is the X matrix corresponding to equation (2), the cluster-level model?

Figure 1. Study design



SWD Model

- X is a $(K \times T) \times (T + 1)$ matrix, where

$$X = \begin{pmatrix} X_{1,1}^T \\ X_{1,2}^T \\ \vdots \\ X_{1,T}^T \\ X_{2,1}^T \\ \vdots \\ X_{K,T}^T \end{pmatrix}$$

and $X_{k,t}$ includes the treatment assignment at time t of cluster k.

SWD Model

- For MaxART, schematic design from slide 5 can be simplified to

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

where there are $T = 9$ time points and 2 clusters per each of 7 patterns (total 14 clusters). To generate the $\boldsymbol{X}$ matrix for the cluster-level analysis, we create the $K \times T$ rows of the design matrix, $\boldsymbol{X}^{\boldsymbol{T}}_{\boldsymbol{k,t}}$, for each cluster at each time t, each of which is of dimension $1 \times (T + 1)$.

SWD model

In MaxArt , cluster (clinic) 6 is in Step 1 pattern. At step time 1 for this cluster, we have

$$\boldsymbol{X_{6,1}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^T$$

Because cluster 6 at time 1 is in standard of care, the second column in $\boldsymbol{X}$ is 0 as are all the time effects for these m_T participants in cluster 6 entering at time 1.
Cluster 6 transitions to the intervention at time 2. Hence,

$$\boldsymbol{X_{6,2}} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^T$$

And at the last time, $T = 9$ the mean of cluster 6 at time 9, or for individual-level analysis, the m_T participants in cluster 6 at time 9, have the following design matrix

$$\boldsymbol{X_{6,9}} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}^T$$

and so on.
Here is another example. Cluster 1 is in the Step 2 pattern. Thus, we have

$$\boldsymbol{X_{1,1}} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^T$$

and

$$\boldsymbol{X_{1,2}} = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^T$$

because cluster 1 at time 1 and time 2 are in standard of care.
At $T = 3$, clinic 1 transitions to the intervention, so cluster 1's mean at time 3, or equivalently for individual level analysis, m_T participants in cluster 1, have the following design matrix

$$\boldsymbol{X_{1,3}} = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}^T .$$

SWD Model

- From (2), the parameter vector is $\boldsymbol{\beta} = (\beta_0, \beta^*, \gamma_1, ..., \gamma_{T-1})^T$
- $\hat{\boldsymbol{\beta}} = (X^T V^{-1} X)^{-1}(X^T V^{-1} Y)$, where $\boldsymbol{X}$ is $(K \times T)x(T+1)$ design matrix for equation (2), the cluster-level model, as given in the previous slide.
- The covariance matrix of $\hat{\boldsymbol{\beta}}$ is $(X^T V^{-1} X)^{-1}$.
- V is now a block diagonal $((K \times T) \times (K \times T))$, each $T \times T$ block on the main diagonal having the standard compound symmetry form for a within-cluster covariance,

$$\begin{pmatrix} \sigma_B^2 + \frac{\sigma_W^2}{m_T} & \sigma_B^2 & \cdots & \sigma_B^2 \\ \sigma_B^2 & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ \sigma_B^2 & \cdots & \sigma_B^2 & \sigma_B^2 + \frac{\sigma_W^2}{m_T} \end{pmatrix}$$

with the $T \times T$ off-diagonal blocks being $\mathbf{0}$ due to independence of clusters.

SWD Model

- We start from equation (8) of Hussey and Hughes (2017):*

$$Var(\hat{\beta}^*) = \frac{K\frac{\sigma_W^2}{m}(\frac{\sigma_W^2}{m} + T\sigma_B^2)}{(KU - W)\frac{\sigma_W^2}{m} + (U^2 + KTU - TW - KQ)\sigma_B^2},$$

where

$$U = \sum_{k=1}^{K}\sum_{t=1}^{T} X_{kt},$$

$$W = \sum_{t=1}^{T}(\sum_{k=1}^{K} X_{kt})^2,$$

and

$$Q = \sum_{k=1}^{K}(\sum_{t=1}^{T} X_{kt})^2$$

*Hussey, Michael A., and James P. Hughes. "Design and analysis of stepped wedge cluster randomized trials." Contemporary clinical trials 28.2 (2007): 182-191.

SWD Model

- Then according to Zhou et al.[†], $Var(\hat{\beta}^*)$ can be simplified to

$$Var(\hat{\beta}^*) = \frac{4\sigma^2}{m_T TK} \frac{(T-1)\{1+(m_T T-1)\rho\}}{(T-2)\{\frac{2}{3}+\frac{1}{3}m_T(T+1)\frac{\rho}{1-\rho}\}}$$

  for models with time effect $\gamma_1, ...\gamma_{T-1}$, where $\sigma^2 = Var(Y_{ktj})$.

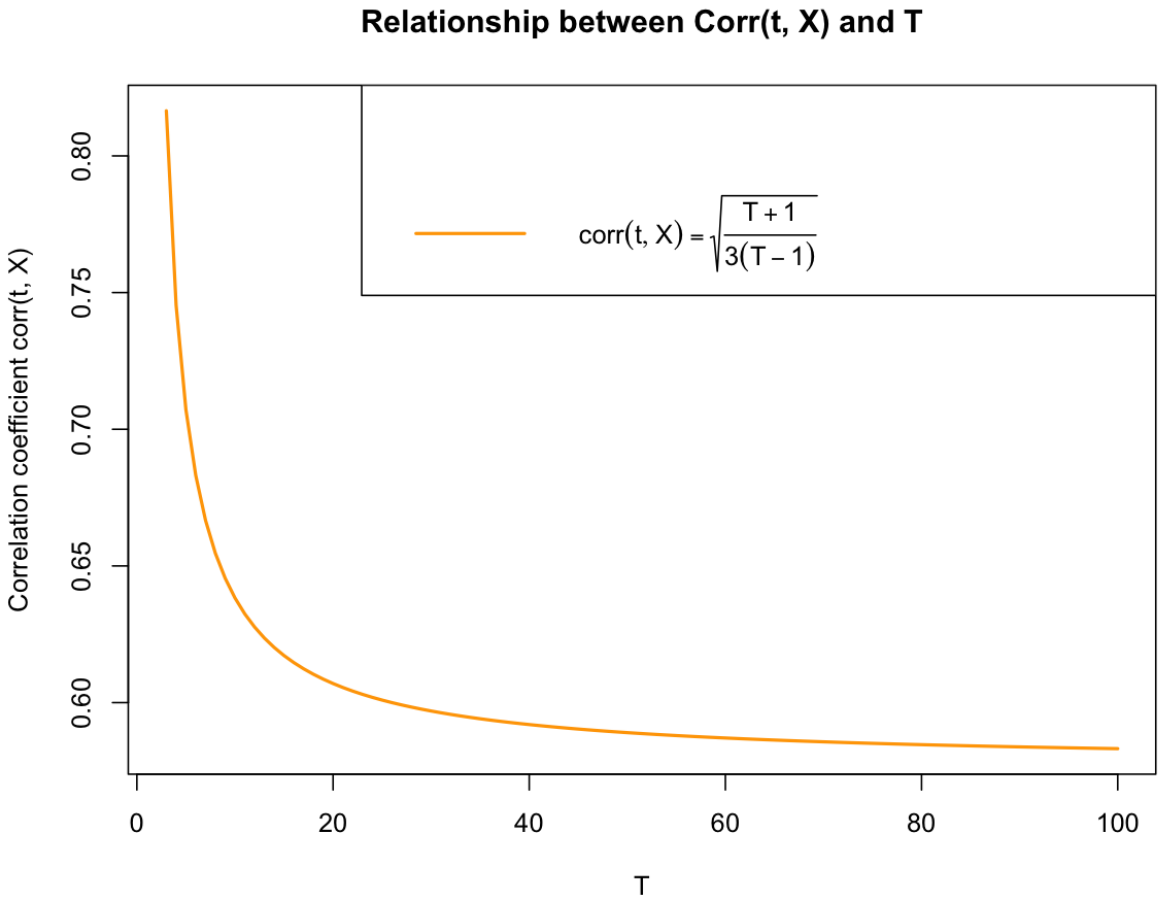
- If there is no time effect,

$$Var(\hat{\beta}^*) = \frac{4\sigma^2}{m_T TK} \frac{1+(m_T T-1)\rho\}}{1+\frac{2}{3}m_T(T+1)\frac{\rho}{1-\rho})}$$

[†] Zhou X, Liao X, Spiegelman D. "Cross-sectional" stepped wedge designs always reduce the required sample size when there is no time effect. J Clin Epidemiol. 2017 Mar;83:108-109. doi: 10.1016/j.jclinepi.2016.12.011. Epub 2017 Jan 16

Correlation between the intervention and time in a SWD

It is shown in the appendix of Zhou (2017) that in a standard SWD, $corr(t, X) = \frac{cov(t,X)}{\sqrt{Var(t)Var(X)}} = \sqrt{\frac{T+1}{3(T-1)}}$



which is less than or equal to 0.82, at its maximum at $T = 3$, the smallest value of T for a SWD, and always greater than 0.58 as $T \to \infty$.

Proof of Corr(X,T) asymptote as $T \to \infty$

Rewriting the formula:

$$\text{corr}(t, X) = \sqrt{\frac{T+1}{3(T-1)}} = \sqrt{\frac{T-1}{3T-3} + \frac{2}{3T-3}} = \sqrt{\frac{1}{3} + \frac{2}{3T-3}}.$$

As $T \to \infty$, the term $\frac{2}{3T-3} \to 0$. Then,

$$\text{corr}(t, X) \to \sqrt{\frac{1}{3}}.$$

Thus, the minimum value of the correlation as $T \to \infty$ is:

$$\text{Minimum corr}(t, X) = \sqrt{\frac{1}{3}} = \frac{1}{\sqrt{3}} = 0.5774 \approx 0.58.$$